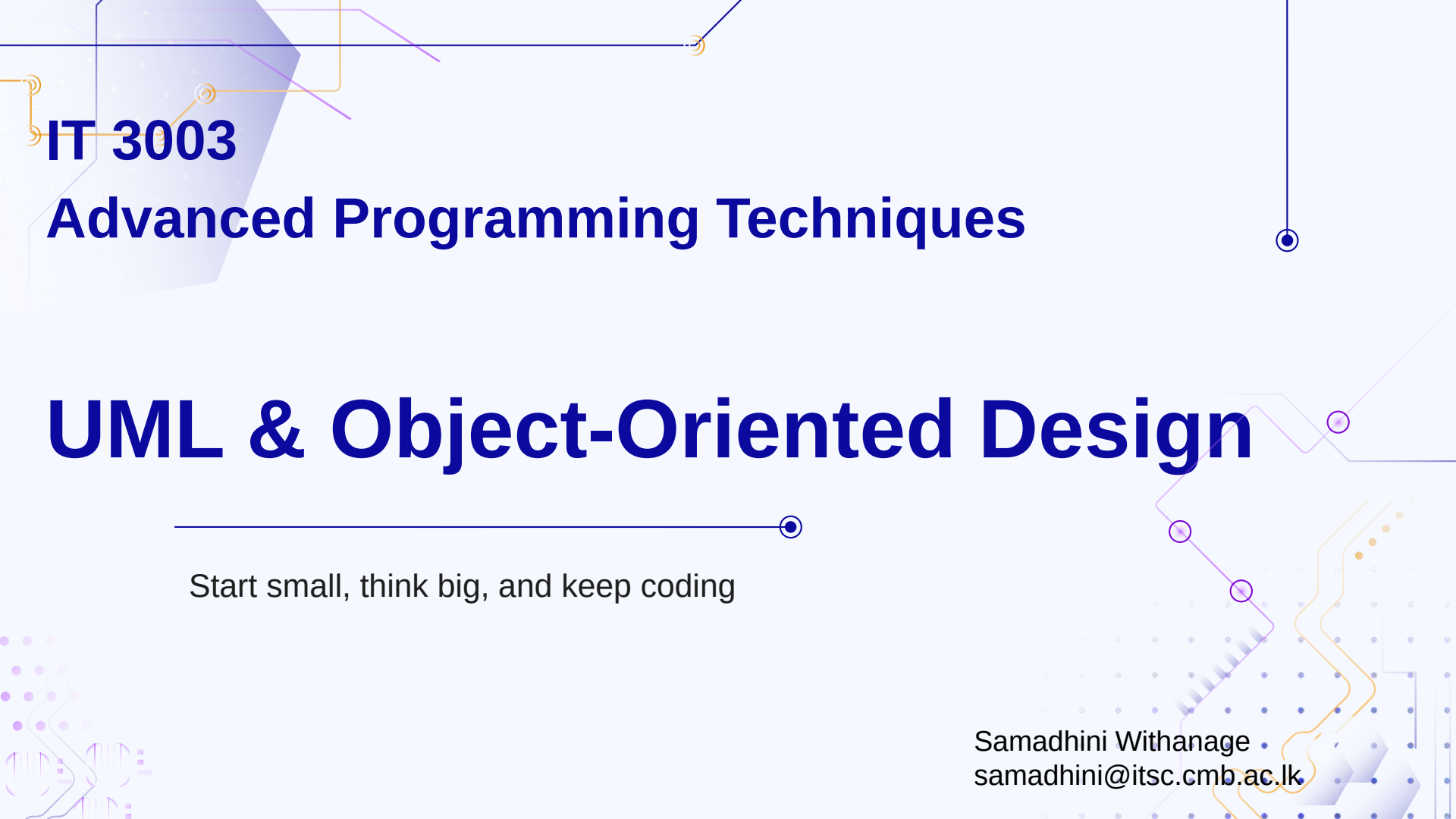




IT 3003

Advanced Programming Techniques

UML & Object-Oriented Design

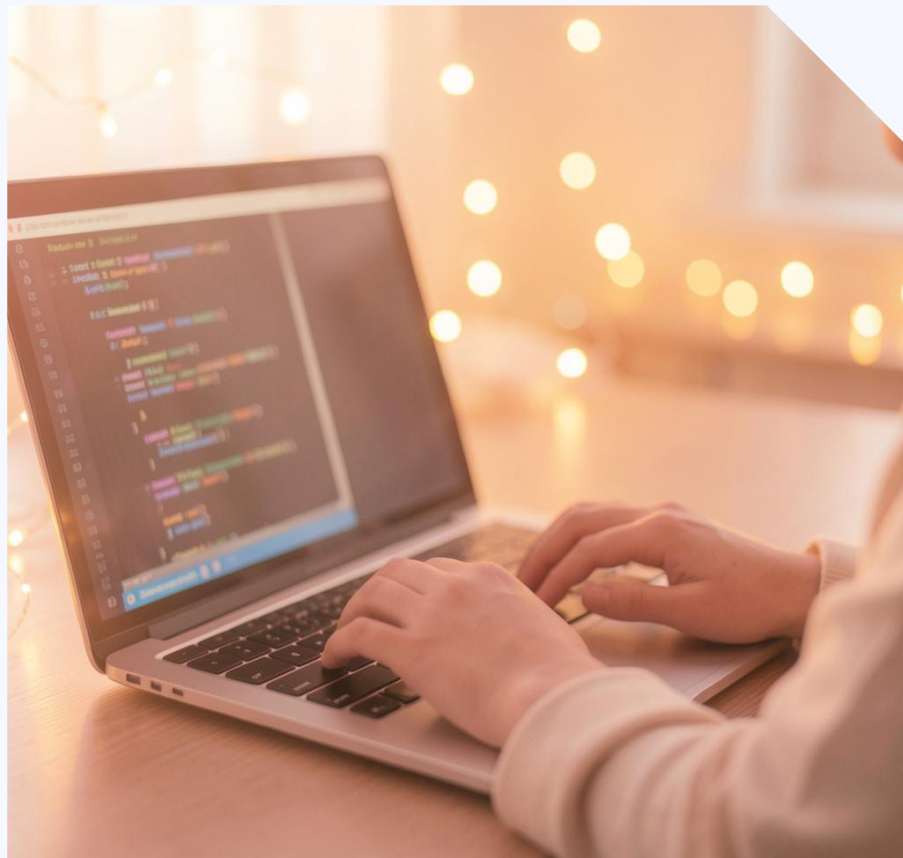
Start small, think big, and keep coding

Samadhini Withanage
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Introduction

Programming is the art of giving clear and logical instructions to a computer to solve problems step by step.

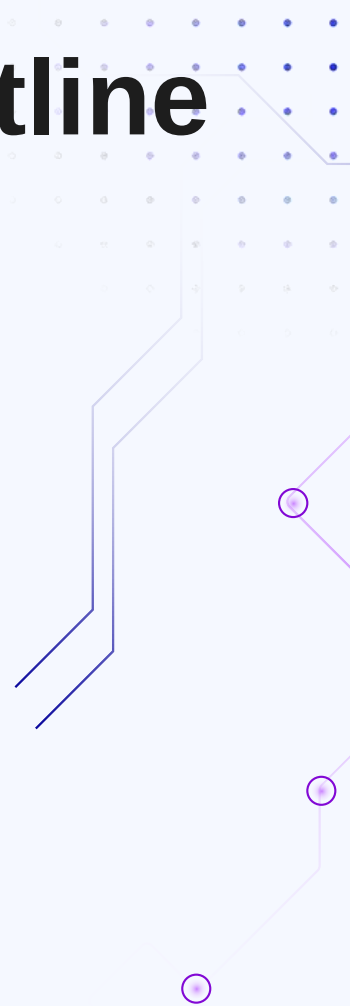
Begin with simple concepts, focus on understanding the logic, and build gradually. Always practice, experiment, and refine your code to develop strong and confident programming skills.





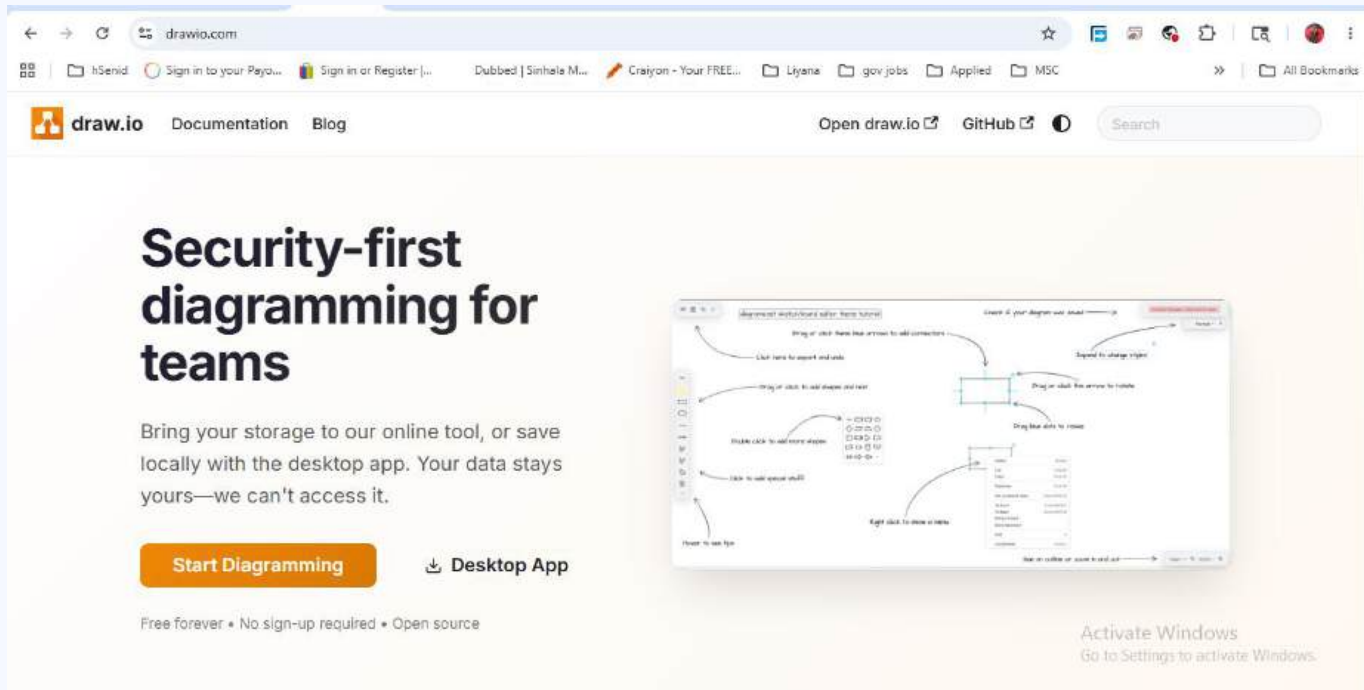
Lesson Outline

- ☀ Tools
- ☀ Class Diagrams
- ☀ Use Case Diagram
- ☀
- ☀



Tools

Draw.io



The screenshot shows the draw.io website homepage. The browser's address bar displays 'drawio.com'. The website's navigation bar includes the 'draw.io' logo, links to 'Documentation' and 'Blog', and buttons for 'Open draw.io', 'GitHub', and a search bar. The main content area features the heading 'Security-first diagramming for teams'. Below this, a paragraph states: 'Bring your storage to our online tool, or save locally with the desktop app. Your data stays yours—we can't access it.' There are two primary call-to-action buttons: 'Start Diagramming' and 'Desktop App'. At the bottom, it says 'Free forever • No sign-up required • Open source'. On the right side, there is a preview of the draw.io interface with various annotations like 'Drag or click these blue arrows to add connections', 'Click here to export and save', 'Drag or click to add shapes and text', 'Double click to add more shapes', 'Click to add special shapes', 'Right click to show context menu', 'Drag or click these blue arrows to add connections', 'Drag or click these blue arrows to delete', 'Drag blue arrows to connect', and 'Click on outline or context menu and...'. In the bottom right corner, there is a message: 'Activate Windows Go to Settings to activate Windows.'

Security-first diagramming for teams

Bring your storage to our online tool, or save locally with the desktop app. Your data stays yours—we can't access it.

[Start Diagramming](#) [Desktop App](#)

Free forever • No sign-up required • Open source

Class Diagrams

Class Creation

Design a class named **Student**.

Attributes

studentId : String

name : String

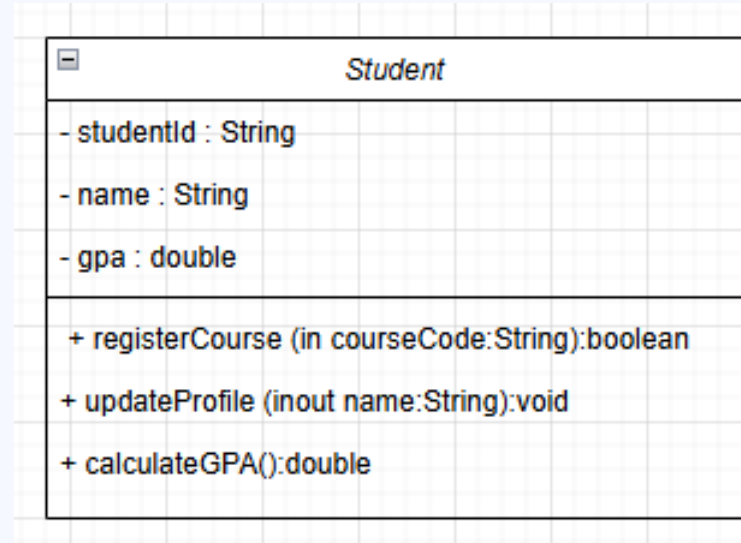
gpa : double

Methods

registerCourse(in courseCode : String) : boolean

updateProfile(inout name : String) : void

calculateGPA() : double



Class Diagrams

Class Creation

Design a class named **BankAccount**.

Attributes

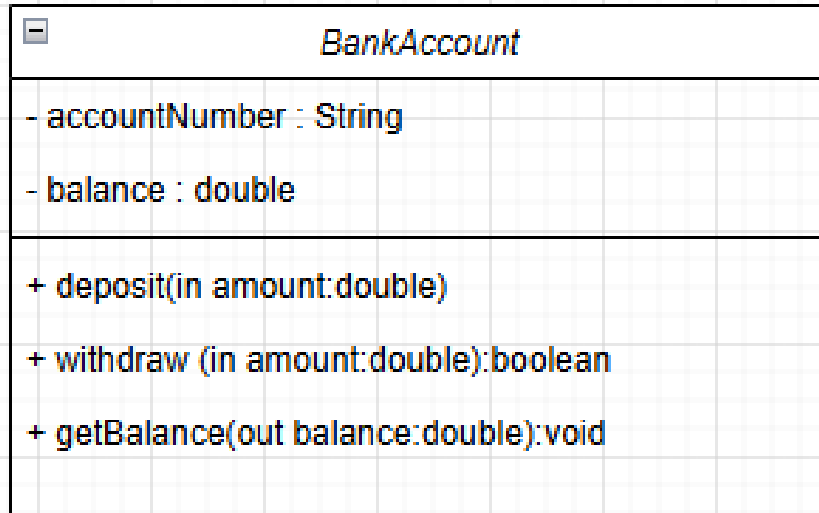
accountNumber : String
balance : double

Methods

deposit(in amount : double) : void
withdraw(in amount : double) : boolean
getBalance(out balance : double) : void

Class Diagrams

Class Creation



Class Diagrams

Class Creation

Design a class named **Employee**.

Attributes

employeeId : String

name : String

salary : double

Methods

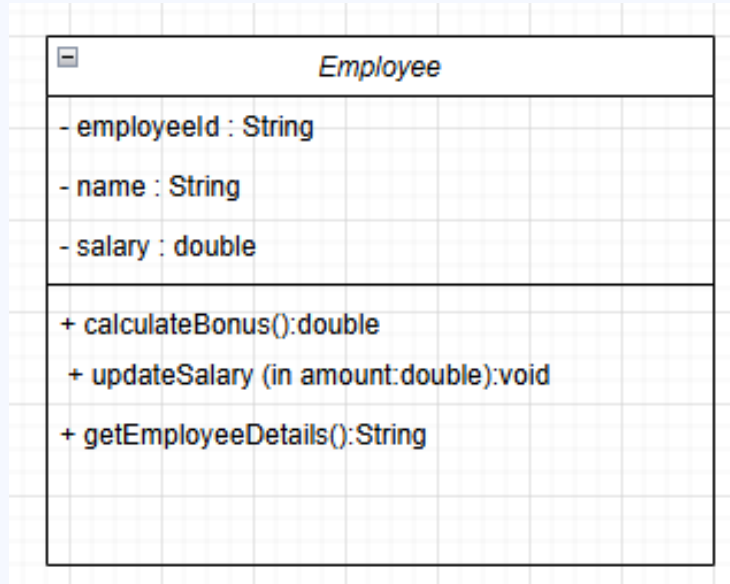
calculateBonus() : double

updateSalary(in amount : double) : void

getEmployeeDetails() : String

Class Diagrams

Class Creation



Class Diagrams

Relationships

1. A doctor treats patients. Patients can continue to exist without the doctor, and doctors can treat many patients
2. A passenger books a flight ticket.
3. A school consists of classrooms. Classrooms can still exist even if the school is closed.
4. A computer contains a motherboard. Without the computer, the motherboard instance in the model does not exist.
5. A savings account is a type of bank account.

Class Diagrams

Relationships

1. Association (Both classes exist independently and are connected through a regular relationship.)
2. Directed Association (The passenger knows about the ticket, but the ticket does not necessarily know about the passenger.)
3. Aggregation (Classrooms can exist independently of the school.)
4. Composition (The motherboard is a part of the computer and its lifecycle depends on the computer.)
5. Generalization (Inheritance) (SavingsAccount "is a" BankAccount.)

Doctor ----- Patient

Passenger -----> FlightTicket

School ◇----- Classroom

Computer ◆----- Motherboard

SavingsAccount -----> BankAccount

Class Diagrams

Cardinality / Multiplicity Practice

1. A person has one national identity card.
2. A customer can place many orders, but each order belongs to only one customer.
3. student can courses
4. A company has several departments. Each department belongs to one company.
5. A lecturer teaches at least one subject and may teach many subjects. A subject may be taught by many lecturers.

Class Diagrams

Cardinality / Multiplicity Practice

1. A person has exactly one national identity card, and each identity card belongs to exactly one person.

One person → One identity card
One identity card → One person

Person 1 ----- 1 IdentityCard

1. A customer may place many orders.
Each order is created by one customer.

Customer 1 ----- 0..* Order

3. A student can enroll in many courses, and a course can have many students.

Many students per course.
Many courses per student.

Student * ----- * Course

Class Diagrams

Cardinality / Multiplicity Practice

4. One company manages many departments.
Each department belongs to one company.
5. A lecturer must teach at least one subject.
A subject can have multiple lecturers.

Company 1 ----- 1..* Department

Lecturer 1..* ----- * Subject

Class Diagrams

Library Management System

A library manages a collection of books. Each book has a book ID, title, author, and availability status.

Library members can borrow books. Each member has a member ID, name, and email address.

A member can borrow multiple books, but each borrowed book is associated with one member at a time.

The library keeps track of all registered members and books.

Class Diagrams

Question 01 - University Course Registration System

A university consists of multiple departments.

Each department employs lecturers and manages students.

Lecturers teach courses offered by their department.

Students can enroll in multiple courses.

Each course contains assignments.

Students submit assignments, and lecturers evaluate them.

The university sends notifications to students regarding assignments and course updates.

Notification services can be implemented using EmailNotification or SMSNotification.

Class Diagrams

Question 02 - ATM Banking System

ATMs are used by customers to perform banking transactions such as withdrawing cash, checking balances, and depositing money.

Each customer owns one or more bank accounts. A customer inserts a bank card into the ATM and enters a PIN for verification.

The ATM communicates with the bank system to validate customer details and process transactions.

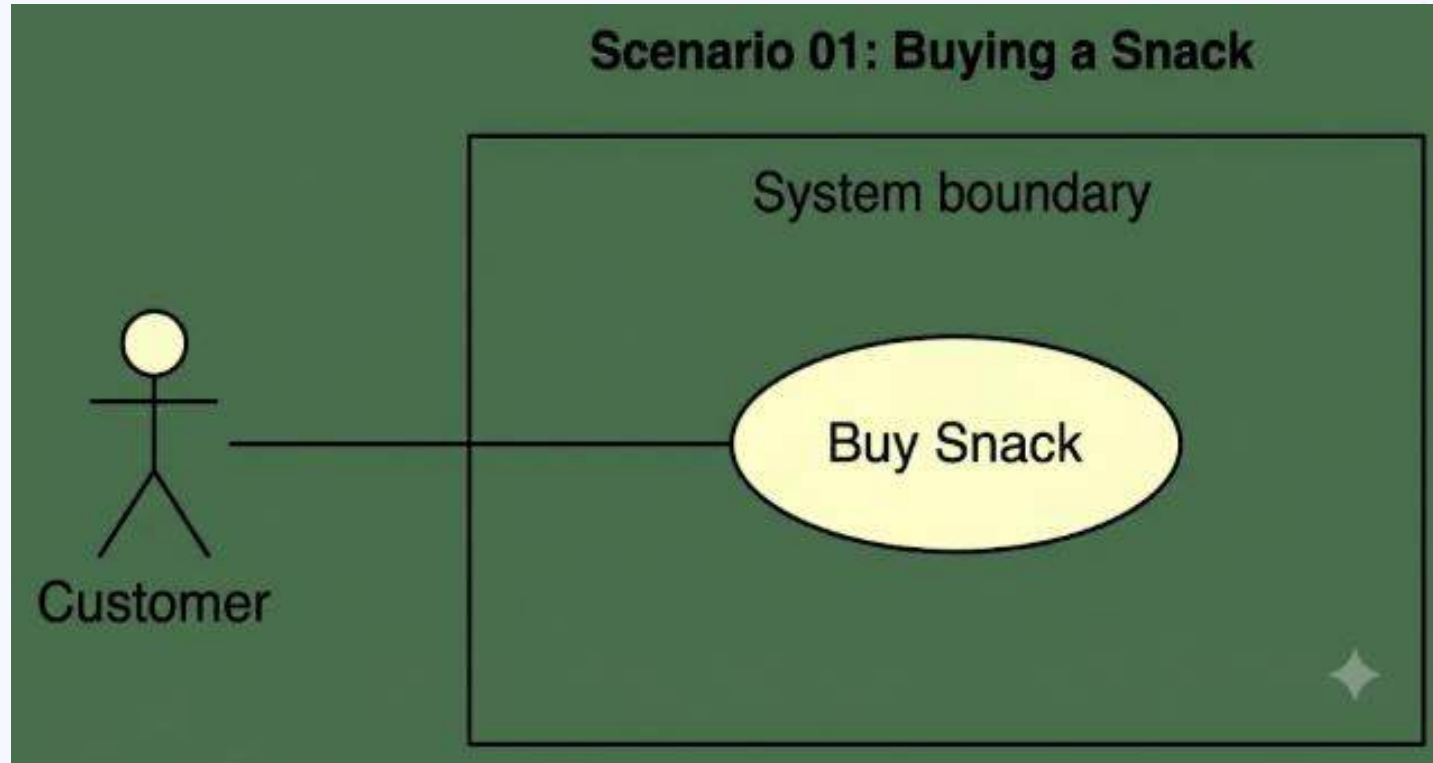
Each transaction records the date, amount, and transaction type.

Use case Diagram

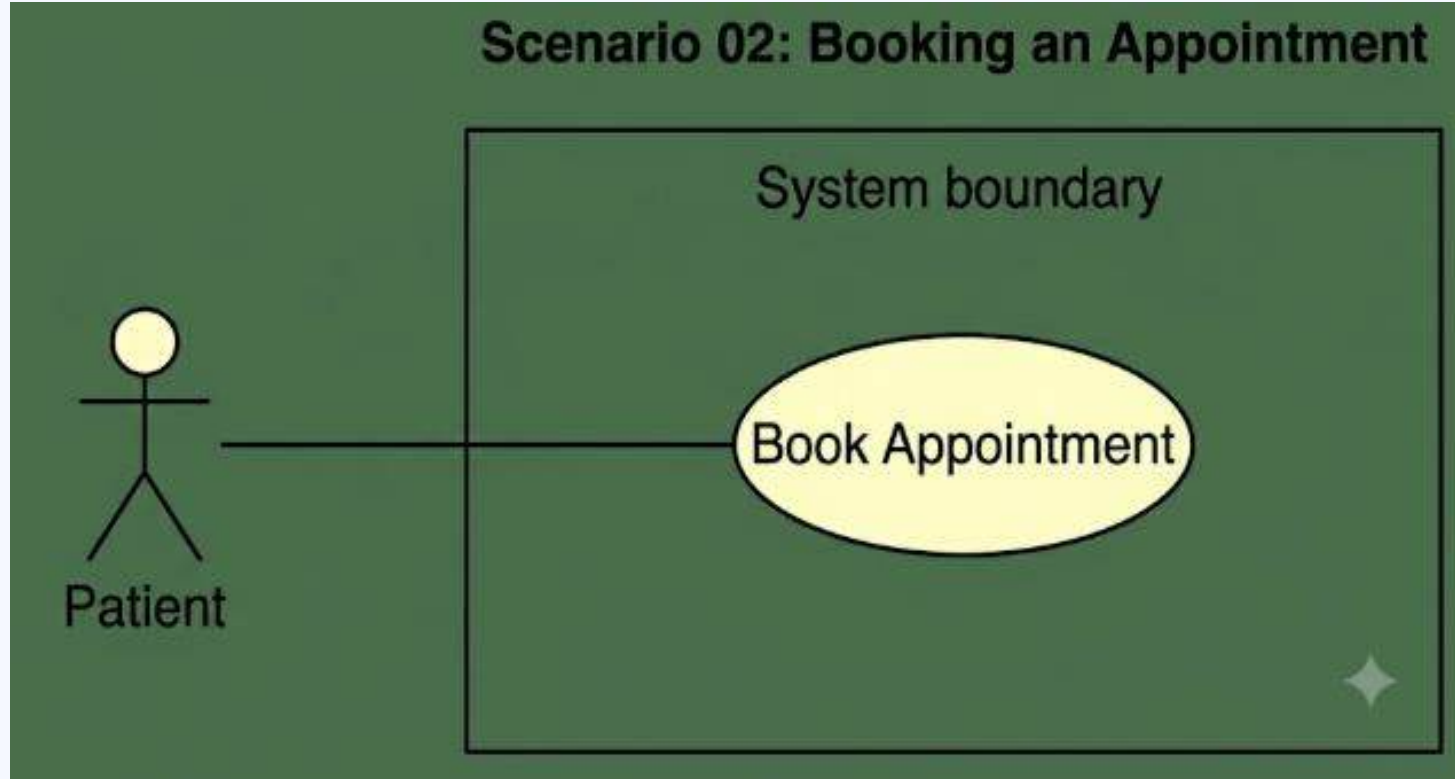
Relationship

1. A customer uses a vending machine to buy a snack.
2. A patient books an appointment through a hospital system.
3. A user logs into an application. The system must validate the username and password before login.
4. A passenger books a train ticket. The system must check seat availability before confirming the booking.
5. A customer purchases a mobile phone online. The customer may optionally purchase an extended warranty.
6. A student submits an assignment. The student may optionally upload supporting documents.
7. An online shopping system has two types of users: Customers and Administrators.

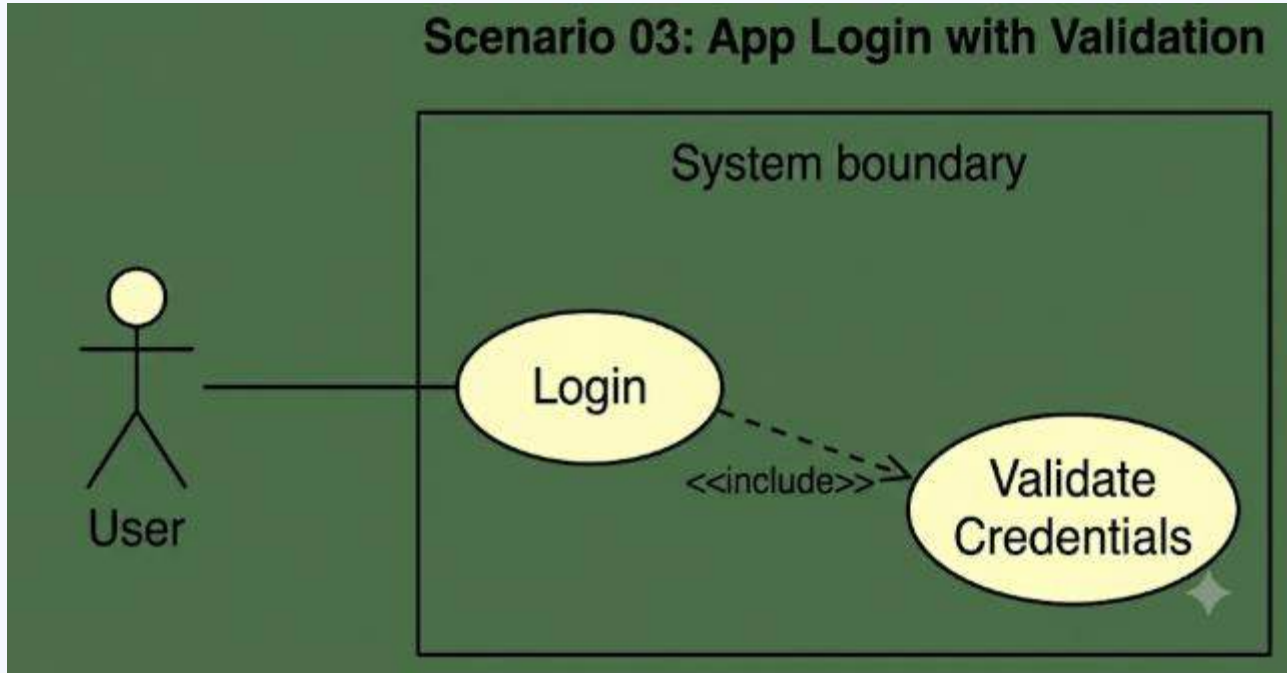
Use case Diagram cont...



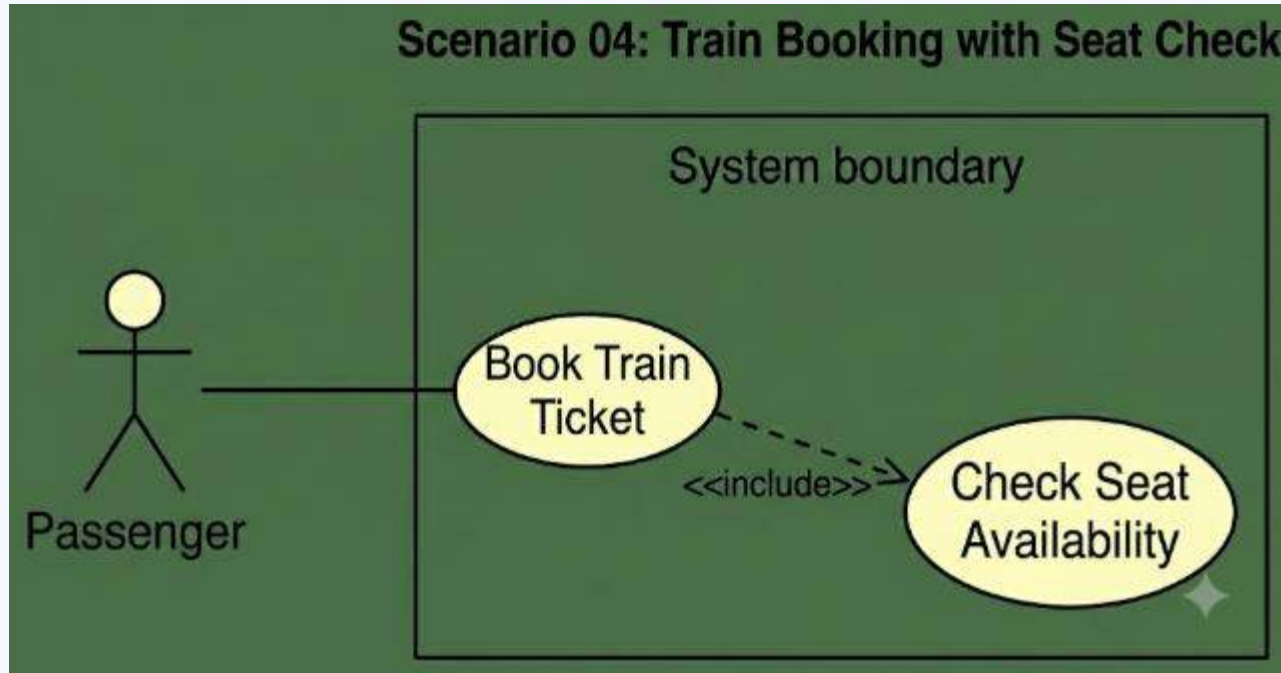
Use case Diagram cont...



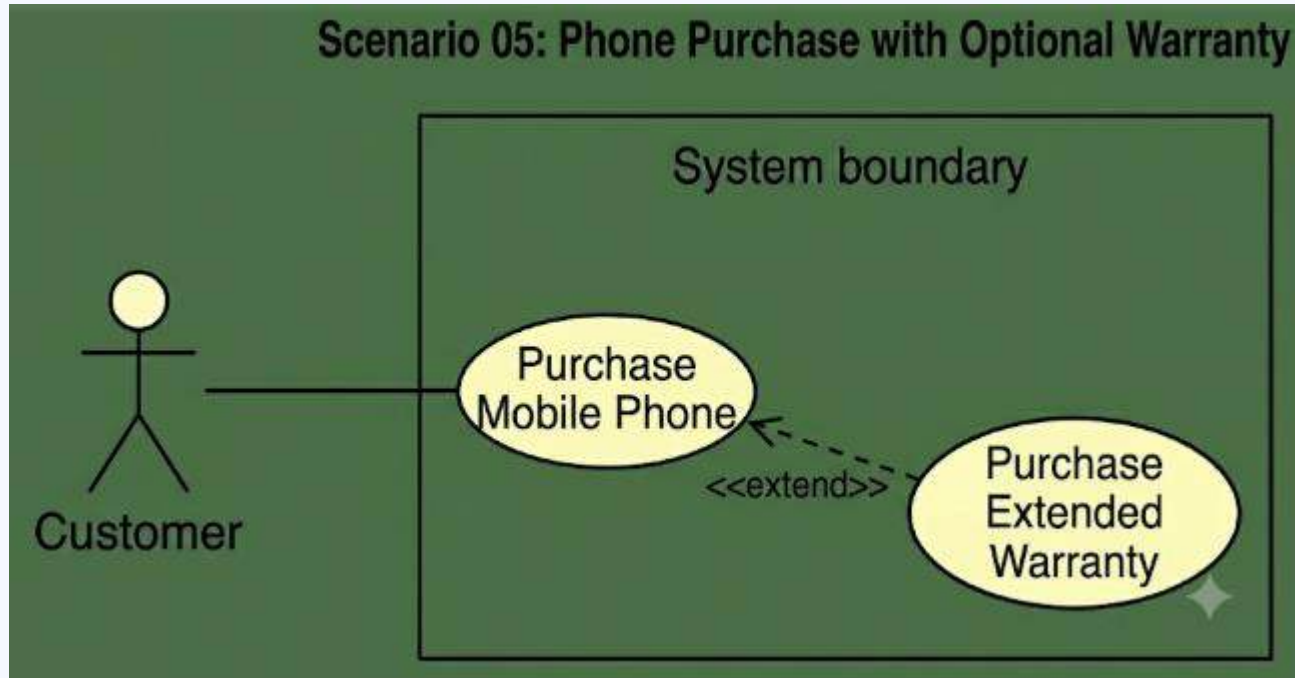
Use case Diagram cont...



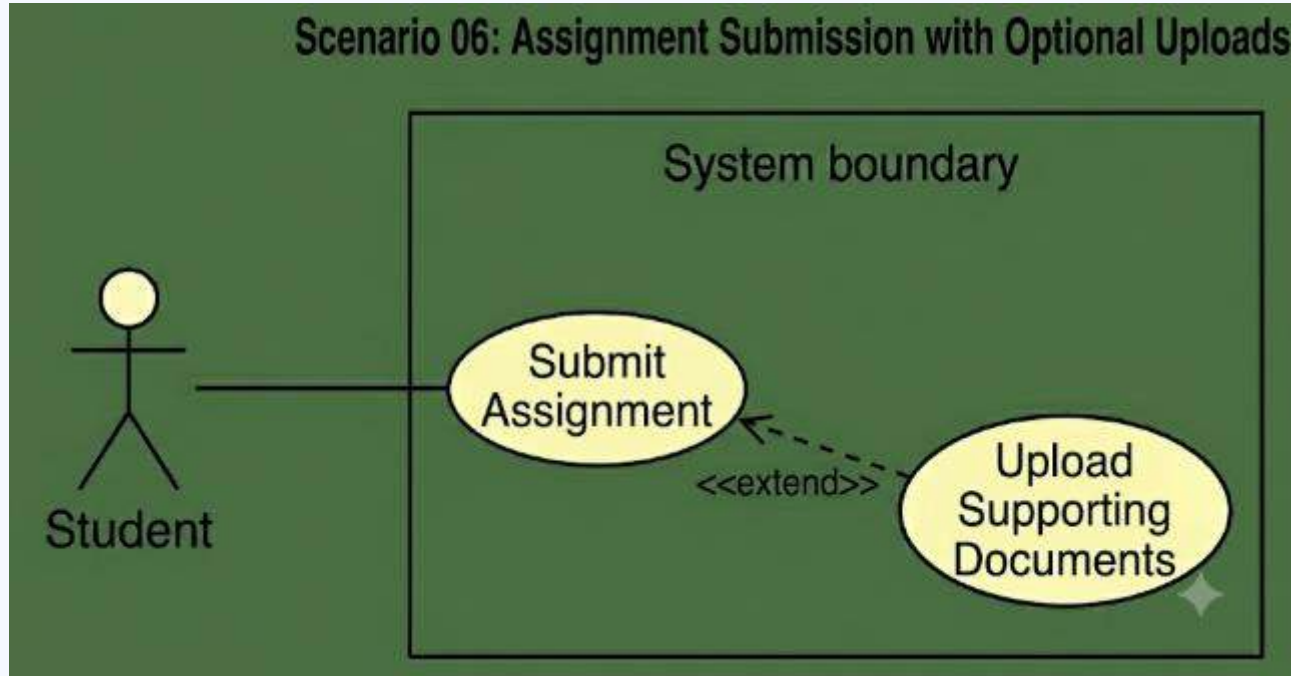
Use case Diagram cont...



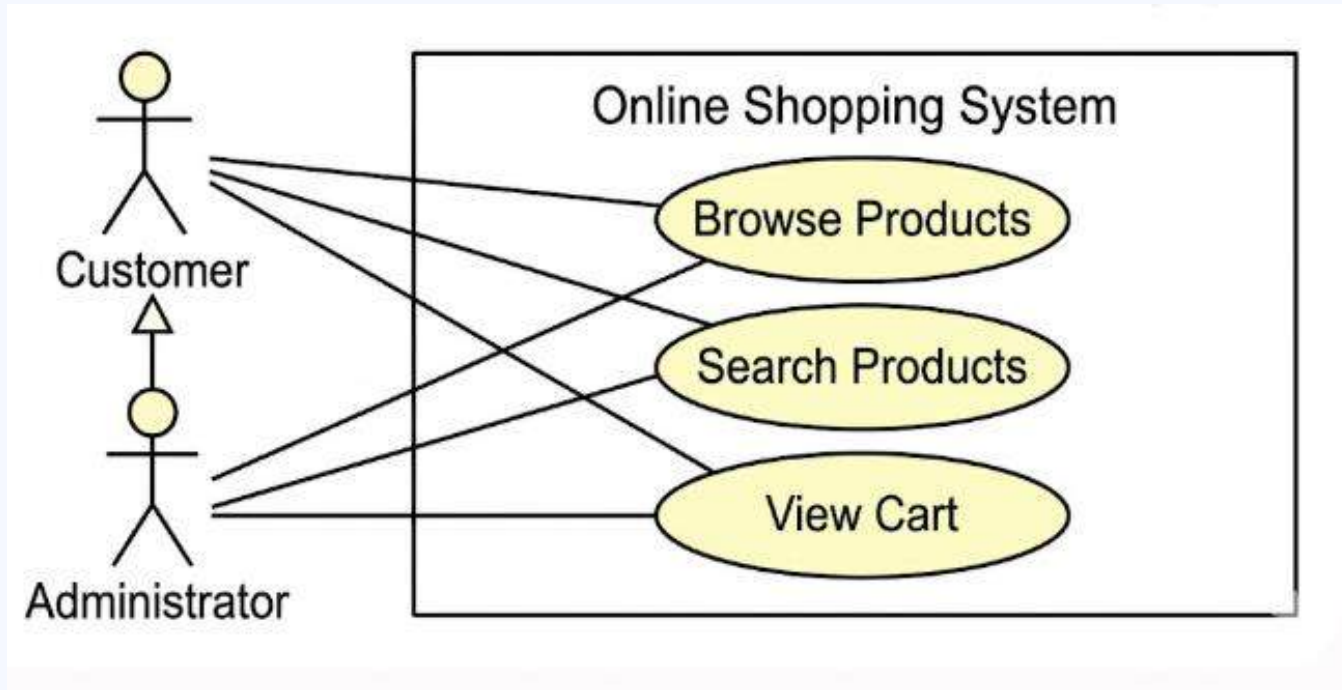
Use case Diagram cont...



Use case Diagram cont...



Use case Diagram cont...



Use case Diagram

University Management System

Scenario

A University Management System is used by Students, Lecturers, and Administrators.

Students can:

- Register for courses
- View course materials
- Submit assignments

Lecturers can:

- Upload course materials
- Create assignments
- View submitted assignments

Administrators can:

- Manage student records
- Manage lecturer records
- Generate reports

Tasks

1. Identify the actors.
2. Identify the use cases.
3. Draw the Use Case Diagram.
4. Show the relationships (Association, Include, and Generalization where appropriate).

Use case Diagram

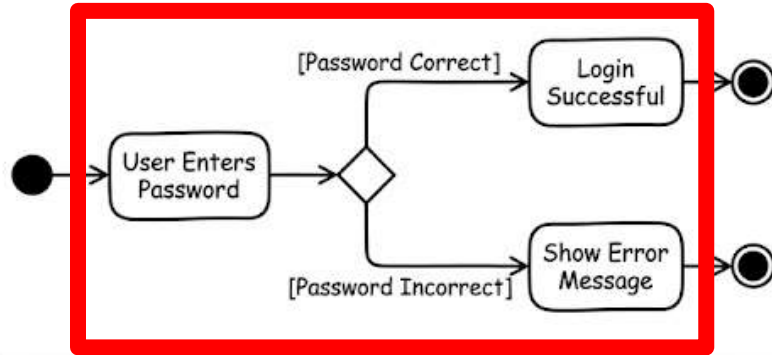


Activity Diagrams

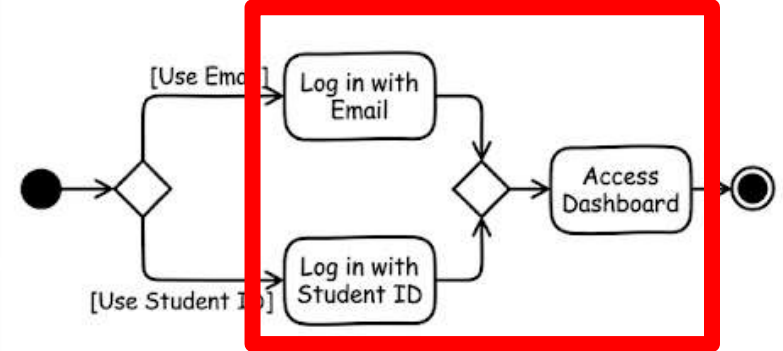
01. A user enters a password.
If the password is correct → Login successful.
If incorrect → Show error message.
02. A student can log in using Email or Student ID.
After either method, access the dashboard.
03. After placing an online order:
Send confirmation email
Update inventory
These happen simultaneously.
04. A university processes student registration.
Verify Payment
Verify Documents
Both must finish before registration approval.

Activity Diagrams cont..

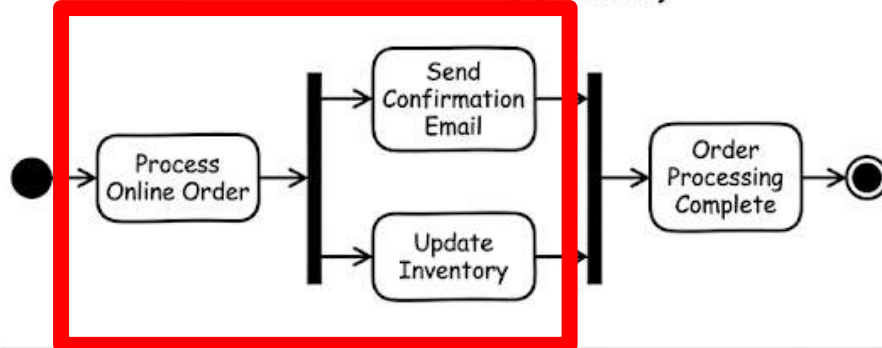
01. Password Check



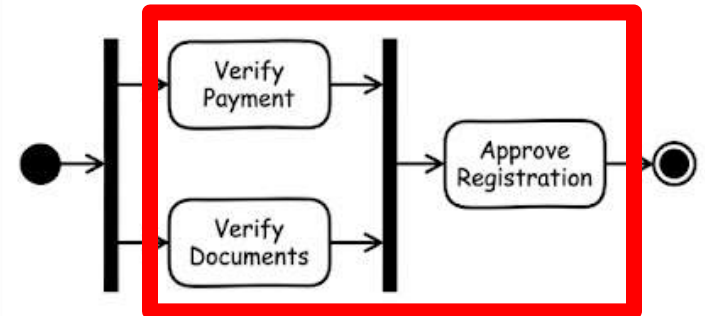
02. Login Options



03. Order Processing (Parallel)



04. Registration Process (Synchronization)



Activity Diagrams cont..

Online Food Ordering Scenario

A customer places a food order.

After placing the order:

- The restaurant prepares the food.

- The payment system processes the payment.

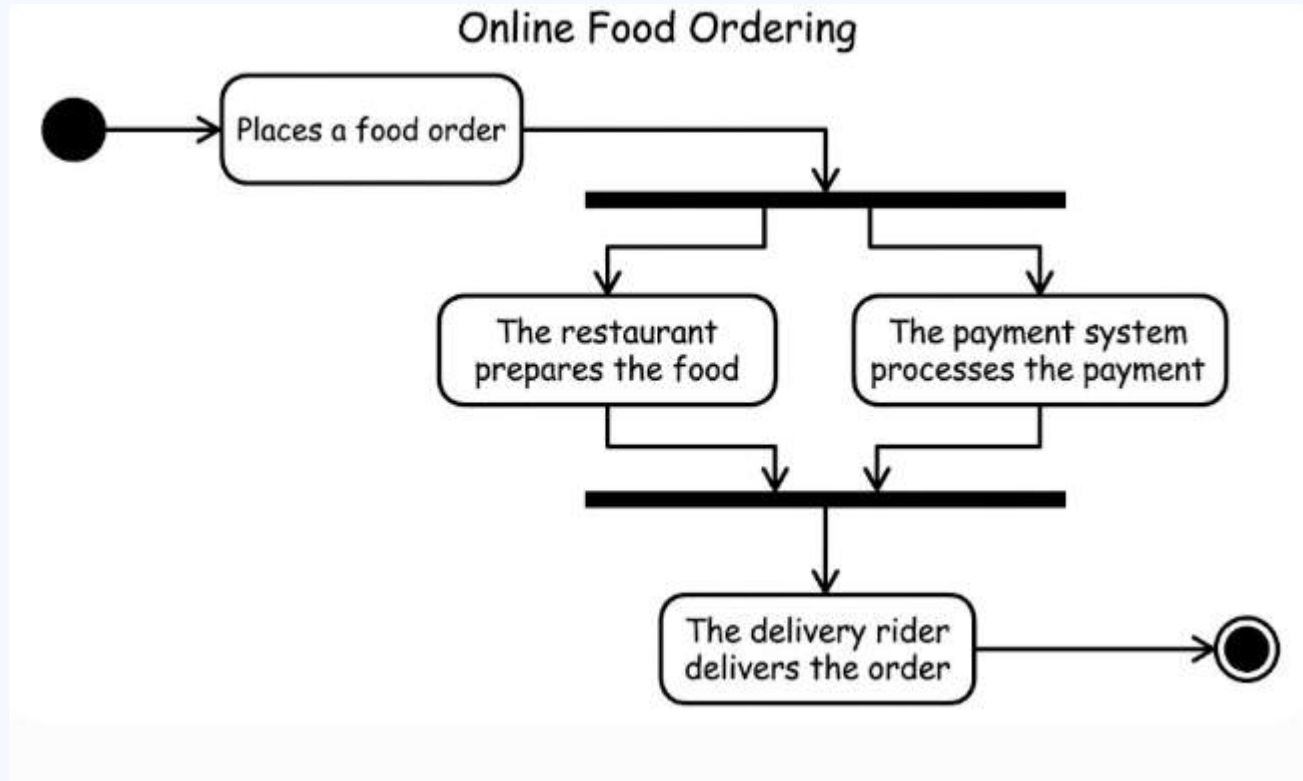
- These activities occur simultaneously.

Once both activities are completed:

- The delivery rider delivers the order.

- The process ends.

Activity Diagrams Cont..



Activity Diagrams cont..

Online Food Ordering Scenario

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After placing the order:

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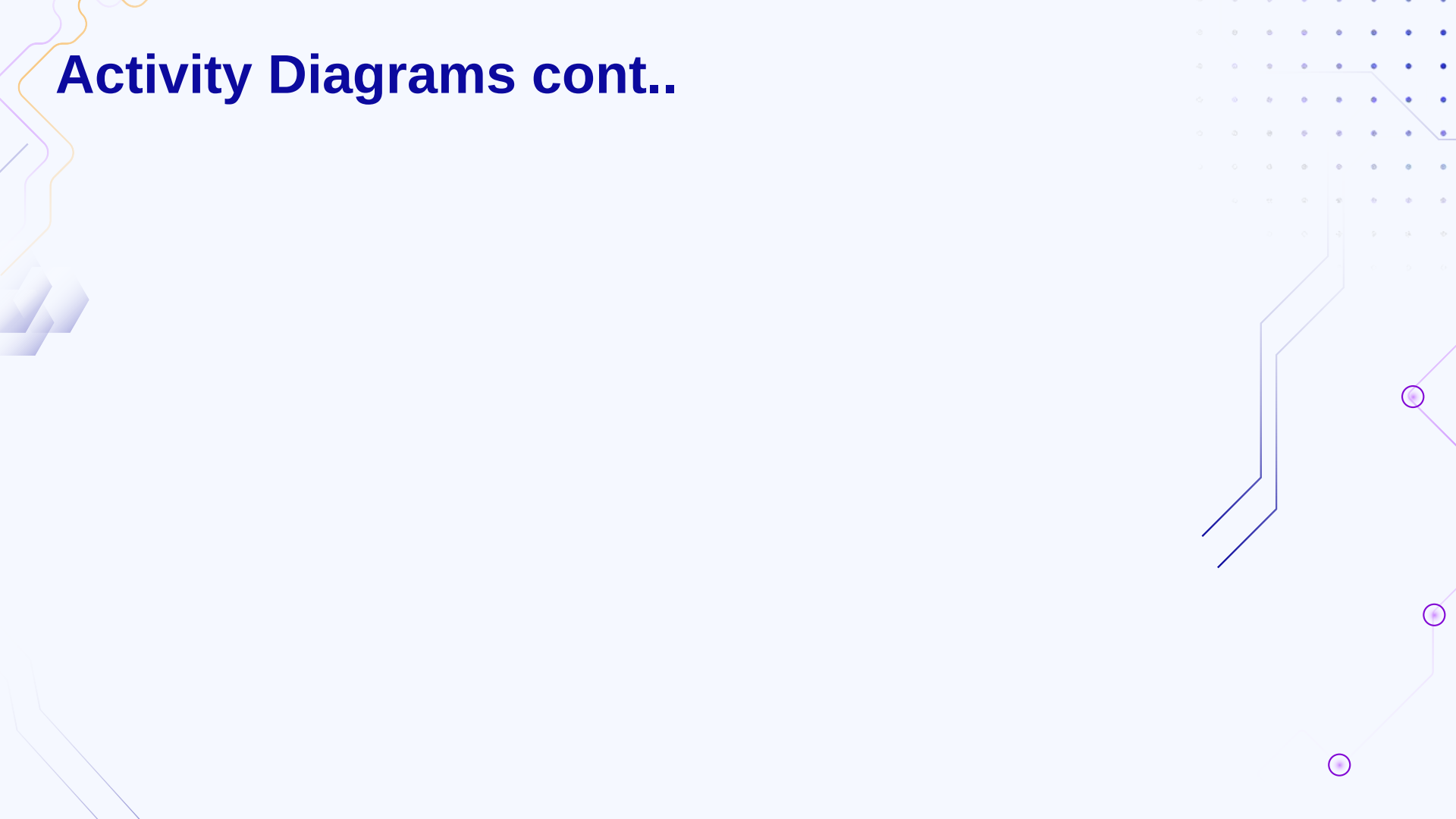
- These activities occur simultaneously.

Once both activities are completed:

- The delivery rider delivers the order.

- The process ends.

Activity Diagrams cont..





Thanks!

